

Code-Layout, Readability and Reusability

Date	1 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Used standard Python indentation (4 spaces) consistently across all modules.
2	Proper File Structure	Files and folders are logically organized	Yes	Project follows a modular architecture with separate folders for models , data , notebooks , src , and app .
3	Meaningful Variable Names	Variables clearly reflect their purpose	Yes	Variables such as user_text , clean_text , emotion_label , tokenizer , label_encoder , prediction_scores clearly indicate their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as preprocess_text() , predict_emotion() , detect_mixed_emotion() , generate_guidance() use meaningful names.
5	Code Comments	Inline and block comments explain logic	Yes	Major modules, preprocessing pipeline, model training, and prediction workflow are documented using comments.

6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules are maintained for preprocessing, BiLSTM, BERT, prediction, training, and Streamlit application.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable functions, common utilities, and modular architecture minimize code duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-except blocks handle model loading, API requests, file operations, and prediction errors.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Text Preprocessing Module	Python, NLTK	Dataset preprocessing, prediction pipeline	High
2	BiLSTM Emotion Model	TensorFlow, Keras	Emotion prediction and training	High
3	BERT Emotion Model	Hugging Face Transformers	Context-aware emotion classification	High
4	AI Guidance Generator	Google Gemini API	Personalized learning response generation	High
5	Visualization Dashboard	Streamlit, Plotly	Emotion analytics and interactive dashboard	Medium

Overall Code Quality Assessment:

S.No	Component / Module Name	Language / Technology
1	Text Preprocessing Module	Python, NLTK
2	BiLSTM Emotion Model	TensorFlow, Keras
3	BERT Emotion Model	Hugging Face Transformers
4	AI Guidance Generator	Google Gemini API
5	Streamlit Dashboard	Streamlit, Plotly